Do the paper's claims match Lean?

Statement audit of `body.tex`

18 October 2026 • 42 Lean source files, 6,632 lines

Verdict

The main mathematical results are substantially aligned. Each of the paper's **13 labelled theorems, lemmas, and corollaries** has a corresponding Lean result, with explicit constants for the bounds hidden by $\lesssim_k$. The backward construction, exact Lanczos output, localization, containment, and the Lipschitz spectral-measure estimate are supported by actual proofs.

The paper does not fully match as written. Its divided Ritz-residual inequality omits a nonzero-vector hypothesis that the Lean theorem explicitly requires. A permitted run can have $V_k y = 0$ for a unit eigenvector y of T_k. The general commutator bound also needs an empty-maximum convention at $k = 1$. The claim that Lean proves "all of the statements in this paper" is too broad for this source tree.

Verification evidence

- `lake build` succeeded: **3,255 jobs**, with linter warnings. The build used the existing local dependency/build cache.
- An executable Lean audit checked the transitive axiom dependencies of **1,156 declarations** under `FinitePrecisionLanczos`, including **890 theorem declarations** (these counts include generated declarations). Only `propext`, `Classical.choice`, and `Quot.sound` occurred.
- A source scan of all 42 project Lean files found no occurrences of `sorry`, `admit`, `axiom`, `unsafe`, `implemented_by`, or `native_decide` as whole words.
- The root import reaches every project module. The backward-model import closure does not include `LossOfOrthogonality`; the Paige proof's import closure does not include the normalized-vector modules.

Findings at a glance

Priority	Finding	Paper location
Required	Add $V_k y \neq 0$ before dividing by $\|V_k y\|$.	Lines 585–594
Required	Restrict the blanket formalization claim to the statements actually covered.	Line 102
Minor	Define $\max \varnothing = 0$ for the nonnegative defect-jump maximum.	Lines 510–515
Clarify	Distinguish Ritz-value localization from normalized Ritz-vector convergence; quantify substantial loss of orthogonality.	Line 596
Scope	Lean assumes a run satisfying a kernel-error model; it does not verify an IEEE implementation or the experiments.	Lines 108–259; 1043–1062

What this report compares

The supplied `body.tex` and `lean/FinitePrecisionLanczos*.lean` source tree are authoritative for this audit. The README was used as a navigation aid, not as proof evidence. The paper and Lean sources were left unchanged. Historical attributions and cited external papers were not independently audited. The referenced `AI` include and experiment assets are absent from this workspace snapshot.

1. A real missing hypothesis

Divided residual: paper lines 585–594

For $x = V_k y$, the paper displays

$$\frac{\|(A-\theta I)x\|}{\|x\|} \le \frac{\sqrt{1+\varepsilon}\,\beta_k |e_k^T y| + \|F_k\|}{\|x\|}$$

without assuming $x \neq 0$. In `Lanczos/Ritz.lean`, line 146, `ritz_residual_bound_div` has the additional hypothesis

```
(hx : 0 < vnorm (L.ritzVector y)).
```

Its undivided counterpart, `ritz_residual_bound` (line 126), needs no such hypothesis. The condition cannot be inferred merely from $\|y\| = 1$: the computed columns may be linearly dependent.

Explicit admissible example

Take $n = 1$, $k = k_{\max} = 2$, $A = [1]$, and $\eta = \varepsilon = 1/1000$. All vector quantities below are scalars. Set

$$v_0 = 0, \quad v_1 = v_2 = 1, \quad v_3 = 0, \qquad \beta_0 = 0, \quad \beta_1 = \eta, \quad \beta_2 = 0.$$

The two steps are

j	u_j	w_j	α_j	z_j	e_j^α
1	1	1	$1-\eta$	η	$-\eta$
2	1	$1-\eta$	$1-\eta$	0	0

All other arithmetic errors and normalization errors are zero. These data satisfy the kernel identities and bounds: in particular $|e_1^\alpha| = \varepsilon\,\|v_1\|\,\|w_1\|$, $\beta_1 v_2 = z_1$, and $2\varepsilon \le 1/100$. The final step is terminal, exactly as allowed by the paper. But

$$T_2 = \begin{pmatrix} 1-\eta & \eta \\ \eta & 1-\eta \end{pmatrix}, \qquad y = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \theta = 1-2\eta, \qquad V_2 y = 0.$$

Thus the displayed ratios and $x/\|x\|$ are undefined in ordinary real analysis. The example's model identities were independently checked with exact rational arithmetic in `check_zero_ritz.py`; it is not a claim about which errors a particular IEEE implementation would produce. Lean's totalized division at zero would not repair the mathematical interpretation.

Recommended change

Before the divided display, insert: "If $x \neq 0$, dividing the undivided residual bound by $\|x\|$ gives…" Keep the undivided inequality as the unconditional statement. Apply the same qualification to the normalized-vector discussion.

The convergence sentence needs care, not a new main theorem

Line 596 identifies a small scalar residual with convergence of the Ritz pair. The proved Paige result is the product bound $s\rho \le C_k \varepsilon\,\|A\|$, where $s = |x^T v_{k+1}|$, $\rho = \beta_k |e_k^T y|$, and $C_k = 3000k^3$. A small ρ alone does not control the divided residual. For a nonterminal step and a specified $s \ge \tau > 0$, however, $\|x\| \ge \tau/\sqrt{1+\varepsilon}$, giving

$$\frac{\|(A-\theta I)x\|}{\|x\|} \le \frac{(1+\varepsilon)C_k \varepsilon\,\|A\|}{\tau^2} + \frac{\sqrt{1+\varepsilon}\,\|F_k\|}{\tau}.$$

This is a mathematical consequence of the checked bounds, not a separate theorem in this project. Alternatively, cite `thm:stabilized` for the conclusion that the *Ritz value* is near the spectrum, which requires no lower bound on $\|x\|$.

2. Assumptions and boundary cases

The common model is consistent

`LanczosRun n k`, defined in `Lanczos/Run.lean`, line 9, stores the matrix, vectors, coefficients, local errors, and their hypotheses. Paper A is Lean `L.H`; paper ε is `L.`ε. The vector norms use Euclidean space, and the matrix norms use `Matrix.Norms.L2Operator`; they are the paper's induced 2-norms, not entrywise norms.

Paper hypothesis	Lean interface	Assessment
$A = A^T$; real data	`hHsymm`; real matrices and vectors	Matches.
$0 \le \varepsilon \le 1/(100k_{\max})$, completed prefix $k \le k_{\max}$	`hε_nonneg`, `hk_pos`, `hsmall`: $k\varepsilon \le 1/100$	Paper implies Lean's prefix condition for $k \ge 1$.
Kernel equations and bounds	`h_u`, `heu`, `h_w`, `hew`, `h_al`, `heal`, `h_z`, `hez`	Matches; $\lvert\beta\rvert$ in Lean equals β by nonnegativity.
Normalization equation and error	`h_be`, `hΔ`	Matches.
Near-unit formed columns; conditional last column	`hnear`, `hnext`	Matches, including terminal case.
$\beta_0 = 0$; internal $\beta_j > 0$; final $\beta_k \ge 0$	`hβ_zero`, `hβ_internal`, `hβ_nonneg`	Matches.
Initial normalization of original input v	No original-input field or normalization routine	Only the near-unit v_1 property is represented.

Termination is handled, with a choice of auxiliary vector

Paper lines 195–259 allow $\beta_k = 0$ and set the unformed $v_{k+1} = 0$. Lean does not require that terminal raw vector to be zero; it is used multiplied by β_k where needed. This weakens the assumptions and causes no conflict for the paper's run.

For the backward construction the paper allows any unit $\bar{v}_{k+1}$ and sets $d_{k+1} = 1$ (line 677). Lean makes the specific permitted choice $\bar{v}_{k+1} = v_1/\|v_1\|$ in `normalizedVec` (`Lanczos/Normalized/Normalization.lean`, line 138). The packaged run theorems therefore certify that choice. The generic dilation algebra is parameterized by matrices and supports the paper's construction, but there is no packaged run-level theorem quantifying over every terminal unit-vector choice. For literal correspondence, specify the same choice in the paper or generalize the run-level interface.

`GreenbaumBackwardModel.exact_next` deliberately assumes $\beta_k > 0$. The theorem unconditionally reproduces T_k, β_k, and the first k exact vectors; it does not identify an auxiliary unit vector with an exact Lanczos output after termination. This agrees with the stated output claim in `thm:greenbaum`. The final sentence of its proof (line 954) should be read with this distinction.

Small indexing clarification in the generic commutator lemma

At $k = 1$, $\max_{1 \le i < k} \beta_i |\gamma_{i+1} - \gamma_i|$ is a maximum over an empty set. The paper gives no convention. Lean's `defectJump` (`Core/SaadGramCommutator.lean`, line 40) inserts zero at the final index and takes a maximum over a nonempty `Fin k`. Consequently this term is zero when $k = 1$. State that convention, and write the lower index limits explicitly. Lean uses $|T_{i,i+1}(\gamma_{i+1} - \gamma_i)|$; this equals the printed expression because the paper's internal β_i are positive. If the lemma is restated for arbitrary signed off-diagonal entries, keep the absolute value on the coefficient. Also make $1 \le k \le k_{\max}$ explicit in the setup.

3. Correspondence: local analysis and Paige

All paths below are relative to `lean/FinitePrecisionLanczos/`. Names in the run-specific rows belong to `FinitePrecisionLanczos.LanczosRun`. "Matches" includes substituting the explicit constants into the paper's $\lesssim_k$ convention, and the assumptions described on page 3.

Paper result	Checked Lean statement and assessment
`lem:coefficients` lines 289–417	`coefficients_paper`, `Lanczos/Bounds.lean`, line 449, proves $$\|u_j\| \le 2\|A\|, \quad \|w_j\| \le 5j\|A\|, \quad \|\alpha_j\| \le 6j\|A\|,$$ $$\|z_j\| \le 7j\|A\|, \qquad \beta_j \le 4j\|A\|.$$ These match the detailed proof as well as the weaker lemma statement. `beta_recurrence_paper` (line 389) proves $\beta_{j+1} \le (1+10\varepsilon)\beta_j + 2\|A\|$. `coefficient_multipliers` (line 10) proves the two multiplier inequalities on the larger interval $[0, 1/20]$. **Matches.**
`thm:local-errors` lines 421–468	`local_errors`, `Lanczos/Bounds.lean`, line 714, proves for $1 \le j \le k$ $$\|f_j\| \le 700j\varepsilon\|A\|, \quad |g_j| \le \varepsilon, \quad |p_j| \le 1200j\varepsilon\|A\|.$$ The error-sum and cancellation identities are proved in `recurrence_error_identity` and `local_product_identity` (`Lanczos/Run.lean`, lines 72 and 93). **Matches.**
`lem:gram-commutator` lines 502–565	`SaadGramRecurrence.gram_commutator`, `Core/SaadGramCommutator.lean`, line 141, proves upper triangularity and $\|N\| \le 2k(P + c\|Z\| + D)$, where P is the maximum adjacent product, c is the maximum norm of the first k columns, and D is the maximum absolute defect jump, padded by zero. The rectangular identity is `padded_commutator` (line 103). Symmetry, tridiagonality, Gram splitting, and the recurrence are explicit structure hypotheses. **Matches after the $k = 1$ convention on page 3.**
`cor:gram-commutator` lines 495–498	`computed_gram_commutator`, `Lanczos/ComputedCommutator.lean`, line 178, supplies the printed matrix definition, the Gram split, upper triangularity, and $\|M_k\| \le 3000k^3\varepsilon\|A\|$. **Matches.**
`thm:paige` lines 612–633	`paige_bound`, `Lanczos/LossOfOrthogonality.lean`, line 41, proves $\|\langle V_k y, \beta_k v_{k+1}\rangle y_k\| \le 3000k^3\varepsilon\|A\|$ for a unit eigenpair. Nonnegative β_k makes this exactly the paper's product. `computed_paige_identity` (line 26) gives the signed identity $\langle V_k y, \beta_k v_{k+1}\rangle y_k = -y^T M_k y$. **Matches.**
`lem:normalized-errors` lines 694–716	`normalized_local_errors`, `Lanczos/Normalized/LocalErrors.lean`, line 144, proves $\|\bar{f}_j\| \le 1500k\varepsilon\|A\|$ and $\|\bar{p}_j\| \le 4800k\varepsilon\|A\|$. The same computed α_j, β_j are used throughout; there is no replacement boundary coefficient. **Matches for the chosen terminal completion.**
`cor:normalized-commutator` lines 732–740	`normalized_gram_commutator`, `Lanczos/Normalized/Commutator.lean`, line 104, proves upper triangularity and $\|\bar{M}_k\| \le 9600k^3\varepsilon\|A\|$ using the normalized recurrence and unit columns. **Matches for the chosen terminal completion.**

4. Correspondence: backward model and Ritz values

Paper result	Checked Lean statement and assessment
`lem:dilation` lines 767–809	`physical_dilation`, `Lanczos/Greenbaum/Statements.lean`, line 23, proves $\widehat{V}_{k+1}^T \widehat{V}_{k+1} = I$ and the prescribed first column. The definitions in `Greenbaum/Dilation.lean` use exactly $k+1$ blocks $\bar{V}_{k+1} C_{k+1} K_{k+1}^r$, $0 \le r \le k$. `triangularC` is a finite geometric sum, with both inverse identities proved in `Core/Nilpotent.lean`; this is the paper's inverse, not a different construction. **Matches for the chosen terminal completion.**
`lem:exchange` lines 824–864	`exchange_rule`, `Lanczos/Greenbaum/Statements.lean`, line 38, proves $\|K_k\| \le 1$, $\|C_k\| \le 2$, $\|L_k\| \le 38400 k^3 \varepsilon \|A\|$, and for each $r \in \mathbb{N}$ an exchange matrix D_r with $\|D_r\| \le r \|L_k\|$. `greenbaumD_zero` and `greenbaumD_succ` (`Greenbaum/Dilation.lean`, lines 201 and 203) prove the printed recurrence for D_r. **Matches.**
`lem:dilated-recurrence` lines 868–904	`dilated_recurrence`, `Lanczos/Greenbaum/Statements.lean`, line 76, proves $\widehat{A}\widehat{V}_k = \widehat{V}_{k+1} \underline{T}_k + \widehat{E}$ with $\left\|\widehat{E}\right\| \le 240000 k^6 \varepsilon \|A\|$. `greenbaumErrorBlock_eq` gives precisely $E_r = \bar{F}_k C_k K_k^r + \bar{V}_k (D_r - D_{r+1})$. The sharper $\sqrt{k}$ and $\sqrt{k+1}$ intermediate estimates are also present in `greenbaumErrorBlock_norm_sqrt` and `greenbaumError_norm_sqrt` (`Greenbaum/Dilation.lean`, lines 241 and 251). **Matches.**
`thm:greenbaum` lines 911–955	`greenbaum_backward_model`, `Lanczos/Greenbaum/Backward.lean`, line 43, proves dimension $n(k+1)$, symmetry, the corrected recurrence, orthonormality, the starting vector, exact output T_k and β_k, and no breakdown before step k. The perturbation satisfies $\|\Delta\| \le 480000 k^6 \varepsilon \|A\|$. `greenbaumDelta_eq` and `greenbaumDelta_projector` (`Greenbaum/Correction.lean`, lines 66 and 54) prove both of the paper's formulas, with $\Delta = \widetilde{A} - \widehat{A}$. `greenbaumDelta_norm` proves $\|\Delta\| \le 2 \left\|\widehat{E}\right\|$. **Matches; terminal output is qualified as on page 3.**
`thm:stabilized` lines 981–1008	`greenbaum_localization_exists`, `Lanczos/Greenbaum/Corollaries.lean`, line 86, proves that an actual eigenvalue λ of A satisfies $$\|\theta - \lambda\| \le \beta_k \|y_k\| + 480000 k^6 \varepsilon \|A\| .$$ The sharper perturbation form and the exact dilated Ritz-residual identity also occur in this file. This is at least as strong as the printed $\lesssim_k$ bound on the sum. **Matches.**
`thm:containment` lines 1013–1039	`greenbaum_containment_eigenvalue`, `Lanczos/Greenbaum/Corollaries.lean`, line 161, proves, for every eigenvalue θ of T_k, $$\lambda_{\min}(A) - 480000 k^6 \varepsilon \|A\| \le \theta \le \lambda_{\max}(A) + 480000 k^6 \varepsilon \|A\| .$$ The code constructs a unit eigenvector for each enumerated eigenvalue; the conclusion is not limited to eigenpairs supplied externally. **Matches.**

5. The spectral-measure remark is supported

`rem:wasserstein`, paper lines 958–964, contains several distinct claims. The code proves their substantive analytic content, including the actual matrix functional calculus and the rank bound.

Eigenvalue localization of the backward matrix

`greenbaumTildeA_eigenvalue_localization` (`Lanczos/Greenbaum/Spectral.lean`, line 37) proves

$$\operatorname{dist}(\widetilde{\lambda}, \operatorname{spec}(A)) \le \|\Delta\|$$

for every eigenvalue of $\widetilde{A}$. This matches the paper's one-sided localization conclusion. The proof uses a symmetric residual estimate, rather than a separately invoked ordered-eigenvalue version of Weyl's theorem. It does not package a multiplicity-preserving matching of all eigenvalues into clusters, which is a different statement from this one-sided bound.

Equality of the unperturbed starting spectral measures

`starting_spectralMass` (`Lanczos/Greenbaum/SpectralMass.lean`, line 14) proves the equality of the finite atomic spectral masses of $(\widehat{A}, \widehat{v})$ and $(A, \bar{v}_1)$. The mass at an eigenvalue is the sum of the squared starting-vector coordinates over its eigenspace, so repeated eigenvalues are combined correctly. `starting_functional_calculus` (`Lanczos/Greenbaum/Wasserstein.lean`, line 28) then proves, for every real function f,

$$\bar{v}_1^T f(A) \bar{v}_1 = \widehat{v}^T f(\widehat{A}) \widehat{v}.$$

Here $f(A)$ is Mathlib's matrix `cfc`, with its eigenbasis interpretation proved in `Core/Measure/FunctionalCalculus.lean`.

The full inequality chain

For every 1-Lipschitz f, the formalization proves

$$\begin{aligned} \left|\bar{v}_1^T f(A) \bar{v}_1 - \widehat{v}^T f(\widetilde{A}) \widehat{v}\right| &\le \left\| f(\widehat{A}) - f(\widetilde{A}) \right\|_{\mathrm{F}} \\ &\le \|\Delta\|_{\mathrm{F}} \le \sqrt{2k}\, \|\Delta\| . \end{aligned}$$

The relevant declarations are:

- `cfc_lipschitz_frobNorm`, `Core/Measure/FunctionalCalculus.lean`, line 82: the full Frobenius-norm Lipschitz inequality for real symmetric matrices.
- `greenbaumDelta_rank`, `Lanczos/Greenbaum/Correction.lean`, line 74: $\operatorname{rank} \Delta \le 2k$.
- `frobNorm_le_sqrt_rank`, `Core/SpectralFrobenius.lean`, line 28: the symmetric rank-to-Frobenius estimate.
- `starting_lipschitz_frobNorm` and `starting_lipschitz`, `Lanczos/Greenbaum/Wasserstein.lean`, line 37 and line 48: the exact scalar bounds printed in the paper.

Meaning of "Wasserstein distance" in this project

`atomicWasserstein` (`Core/Measure/Wasserstein.lean`, line 26) is defined as the supremum of absolute integral differences over all 1-Lipschitz functions. `starting_wasserstein` proves its bound by $\sqrt{2k}\,\|\Delta\|$; `starting_wasserstein_explicit` inserts $480000k^6\varepsilon\,\|A\|$. This matches the paper's explicitly stated Lipschitz-test formulation.

There is no project theorem identifying this finite-atomic dual definition with an infimum over transport couplings or a library probability-measure Wasserstein metric. That is a formalization boundary, not a failure of the displayed Lipschitz inequality. If that identification is intended to be part of the formalization claim, state the dual convention or add the bridge theorem.

6. Coverage limits and recommended wording

The "all statements" claim is not supported literally

Paper line 102 says that formal Lean proofs are supplied for all statements. The source tree supports the numbered analytic results described in this report, subject to the qualifications above, but does not contain all of the following:

- A formal execution semantics for Algorithm 1, an IEEE arithmetic model, or a theorem constructing `LanczosRun` from an execution of that algorithm. The kernel bounds and normalization properties are hypotheses, as the paper's assumption-based analysis intends.
- A theorem relating the given original starting vector to the first computed vector through the initial normalization perturbation (lines 190–193). The run model starts with its stored near-unit v_1.
- A general theorem that exact Lanczos on an arbitrary symmetric input produces an orthonormal Krylov basis, including the span equality in lines 24–30. `Core/ExactLanczos.lean` defines a process and identifies its outputs *given* an orthonormal recurrence; its structure `OrthonormalRecurrence` assumes orthogonality. This is sufficient for the backward theorem, where orthogonality is independently proved by dilation.
- Verification of the numerical values, cluster counts, or weight errors in the figures and numerical illustration (lines 41–44, 603–608, 1043–1062). There is no instantiation of the Lean model with those experiment data.
- Proofs of historical comparisons, claims about practical accuracy, or statements about the precision needed for kernels and downstream algorithms. Those require their own evidence; they are not consequences certified by the current project merely because they occur in the paper.

Suggested replacement for line 102:

> The accompanying Lean development proves counterparts of the main local-error, Gram-commutator, loss-of-orthogonality, backward-model, localization, containment, and Lipschitz spectral-measure results, with explicit constants, for runs satisfying the stated arithmetic model. The implementation-to-model connection, introductory background results, and numerical experiments are not formalized.

Numerical illustrations need separate validation

The formal perturbation theorem is about the exact real matrices in its construction. Forming the model in double precision, as described in the paper, introduces additional errors that the Lean theorem does not bound. Likewise, double precision with full reorthogonalization is a numerical reference for exact Lanczos, not an exact-arithmetic certificate (figure caption, line 42). The observed $4 \cdot 10^{-6}$ weight agreement and $2.5 \cdot 10^{-7}$ perturbation norm should be presented as experimental measurements.

The one-sided localization theorem does not by itself certify that each original eigenvalue has a cluster of exactly $k + 1$ numerical model eigenvalues. Such a count needs a matching/separation argument or direct validated numerical evidence. This audit makes no claim that the reported experiment is incorrect.

Documentation drift

`lean/README.md`, lines 30–31, says the nonzero condition on the divided Ritz residual appears in both paper and code. It is present in the code but absent from this `body.tex`. The README also links to a correspondence map and prior reports not supplied in this workspace. Those assertions were not used to establish the verdict.

Minimum changes to align the paper

Add the nonzero condition, define the empty defect-jump maximum, narrow the formalization claim, and qualify the convergence sentence. For exact statement correspondence at termination, specify the same auxiliary unit-vector choice as Lean. None of these changes requires weakening the proved main stability bounds.

7. Traceability and reproduction

Additional displayed identities checked

Paper equation or proof step	Lean evidence (paths relative to the project namespace)
eq:governing	governing, Lanczos/Ritz.lean:63
eq:gram-split, eq:M-def	computed_gram_split, computedCommutator, Lanczos/ComputedCommutator.lean:31,63; generic padded_commutator
eq:gram-commutator-full	GramRecurrence.gram_commutator_full, Core/GramCommutator.lean:129
eq:generic-commutator	SaadGramRecurrence.N_skew, Core/SaadGramCommutator.lean:112
eq:ritz-residual-identity	ritz_residual_identity, Lanczos/Ritz.lean:92
eq:normalized-interface	normalized_governing_succ, Lanczos/Normalized/Recurrence.lean:47
Normalized error comparison	normalizedF_sub_identity, normalizedP_identity, Lanczos/Normalized/Recurrence.lean:53,67
eq:normalized-gram	normalized_gram_split_succ, normalizedRsucc_block, Lanczos/Greenbaum/Dilation.lean:21,37
eq:CKsy, eq:Kkp-power	greenbaum_CK_blocks, greenbaum_Kkp_power, Lanczos/Greenbaum/Dilation.lean:61,118
eq:CGC-identity	greenbaum_isometry_identity, Lanczos/Greenbaum/Dilation.lean:109
eq:L-commutator	greenbaumL_commutator, Lanczos/Greenbaum/Dilation.lean:169
eq:delta_as_projector	greenbaumDelta_projector, Lanczos/Greenbaum/Correction.lean:54

Reproduce the checks

From lean_audit/lean, run:

```
lake build
lake env lean ../audit_report/AxiomAudit.lean
```

From lean_audit, run:

```
python3 audit_report/check_zero_ritz.py
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=audit_report audit_report/statement_audit.tex
```

The report is standalone; it does not require the paper's missing preamble, figures, bibliography, or AI include.

Evidence retained alongside this PDF

build.log, axiom_audit.log, AxiomAudit.lean, source_scan.txt, source_hashes.json, import_audit.json, check_zero_ritz.py, and zero_ritz_check.log. The axiom audit uses Lean's collectAxioms on every imported declaration with the namespace prefix; private dependencies are covered transitively when used by those declarations. It rejects any axiom outside the stated allowlist.

Snapshot identification and limits

Toolchain: leanprover/lean4:v4.32.2. Mathlib manifest revision: 905b95818eb32af7874a58b427f50c1711a5e96c. The dependency packages are provided by the existing local symlink/cache; this was not a clean bootstrap or an independent rebuild of Mathlib. Paper SHA-256: 18244124f5f6aa17a9324ea298a75aa1959136a390404dc169b9a845d9344e28.

The source manifest records individual hashes for the paper and all 42 Lean files. The prose-to-formula comparison is a human-readable semantic audit, not a mechanized translation of LaTeX into Lean. Kernel acceptance of Lean proofs and correspondence of their statements to the paper are separate checks; this report records both and states the remaining gaps explicitly.